

Applying Machine Learning to Real-World Data Challenges

Inquiry-Based Learning Report

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Assignment	Data Challenge Scenarios
Artifact	Professional Portfolio Artifact 4

This report transforms a guided AI coaching conversation into a professional learning artifact while preserving the scenarios, reasoning, feedback, revisions, and conclusions developed during the inquiry.

1. Introduction

This artifact documents an inquiry-based learning activity focused on practical machine learning challenges. The guided conversation presented three increasingly complex scenarios involving missing data, severe class imbalance, and data or concept drift. For each scenario, I explained my initial approach, received feedback, and revised my reasoning to produce a stronger and more production-oriented solution.

The discussion then expanded into post-deployment monitoring and the emerging challenges of maintaining large language model systems. Together, these topics connected foundational machine learning concepts with business impact, operational reliability, and continuous improvement.

2. Scenario 1: Missing Data

The first scenario involved a customer churn dataset with 10,000 records and 15 features. Three variables contained significant missing values: `last_login_date` at 40%, `customer_support_tickets` at 25%, and `payment_method` at 12%. The goal was to prepare the data for a random forest classifier without losing useful information.

Initial Approach

I began by avoiding immediate imputation. Instead, I first considered why each value was missing, because the missingness itself could contain information about customer behavior.

- For `last_login_date`, I would determine whether missing values meant the customer never logged in or whether the value was simply not recorded.
- For `customer_support_tickets`, I would check whether a missing value could reasonably mean zero support contacts.
- For `payment_method`, I would use the mode or an Unknown category rather than delete records.

Feedback and Revision

The AI coach encouraged me to evaluate whether `last_login_date` was predictive before dropping it, determine whether missingness correlated with churn, and verify that imputation did not distort the feature distribution.

In response, I strengthened the solution by proposing feature-importance analysis, churn-rate comparisons for customers with and without login data, missing-value indicator variables, and distribution checks before and after imputation. I also noted that model validation should be used to compare alternative imputation strategies.

3. Scenario 2: Imbalanced Fraud Detection

The second scenario involved 100,000 financial transactions, but only 800 transactions, or 0.8%, were fraudulent. A false negative cost approximately \$5,000, while a false positive cost only \$50. This created both a class imbalance problem and a strong cost asymmetry.

Modeling Priorities

- Prioritize recall so the model captures as many fraud cases as possible.
- Use class weighting or oversampling methods such as SMOTE to increase attention to the minority class.
- Evaluate precision, recall, F1-score, and the confusion matrix instead of relying on overall accuracy.
- Tune the probability threshold rather than automatically using 0.50.

Cost-Based Threshold Optimization

The coach asked how I would determine the threshold quantitatively. I revised the approach by proposing a validation-set search across multiple thresholds. For each threshold, I would calculate total expected business cost using the

number of false negatives multiplied by \$5,000 plus the number of false positives multiplied by \$50. The selected threshold would be the one that minimized total cost, even if it produced more false positives.

4. Scenario 3: Data Drift and Concept Drift

The third scenario involved a credit-risk model trained on 2018-2019 loan data. After the bank introduced a new digital lending platform aimed at younger borrowers with shorter credit histories and alternative income sources, monitoring showed shifted feature distributions, more high-risk predictions, and actual default rates 2.3% higher than predicted.

Diagnosing the Problem

I first distinguished data drift from concept drift. Data drift refers to changes in the distribution of input features, while concept drift refers to changes in the relationship between those features and the target outcome. In this scenario, both were possible because the customer segment changed and the model began underpredicting default risk.

Quantitative Drift Measurement

- Population Stability Index (PSI) for distribution shifts over time.
- Kolmogorov-Smirnov test for numerical features.
- Chi-square test for categorical features.
- Calibration and discrimination metrics to determine whether predictive performance declined.

Retraining Strategy

The coach challenged the assumption that adding new records to the original training set would automatically solve the problem. I revised the solution by proposing feature-importance analysis for the new segment, alternative features such as gig-income patterns, and separate models if a single model could not perform well across both customer groups. Any revised model would be compared against the current model on a representative validation set before deployment.

5. Comparison of the Three Scenarios

Scenario	Primary Challenge	Core Response	Business Focus
Missing Data	Incomplete features with meaningful missingness	Investigate mechanisms, preserve signal, validate imputation	Avoid bias and information loss
Fraud Detection	0.8% positive class and asymmetric costs	Class weighting, recall, cost-based thresholding	Minimize financial loss
Model Drift	New customer segment and degraded calibration	Drift tests, feature review, retrain or segment models	Maintain reliability over time

6. Best Practices for Production Monitoring

After completing the three scenarios, I asked how to monitor deployed models so they continue performing well. The discussion identified several monitoring layers:

- Model metrics such as precision, recall, F1-score, AUC, or RMSE.
- Business metrics such as fraud losses prevented, conversion rates, and customer impact.

- Operational metrics such as latency, throughput, error rates, and resource consumption.
- Data-quality checks for missing values, schema changes, new categories, and invalid ranges.
- Drift detection through thresholds, statistical tests, and rate-of-change alerts.
- Golden datasets, explainability tools, shadow deployments, A/B testing, human feedback, automated retraining, and model versioning.

7. Monitoring Large Language Model Systems

The final inquiry compared traditional machine learning monitoring with the broader challenges introduced by large language models and generative AI systems.

Monitoring Area	Traditional ML	LLM or Generative AI
Output evaluation	Clear labels and numerical metrics	Quality, factuality, helpfulness, safety, and task-specific evaluation
Reproducibility	Mostly deterministic predictions	Sampling, temperature, and nondeterministic outputs
Inputs	Fixed structured features	Prompts, context, retrieved information, and conversation history
Risk	Prediction error and fairness	Hallucination, unsafe content, prompt injection, and privacy leakage
Operations	Latency and throughput	Token usage, provider changes, tool calls, cost, and modelversion regressions

8. Process Used

1. Analyze the business and technical context of each scenario.
2. Identify the primary machine learning challenge.
3. Propose an initial solution and explain the reasoning.
4. Review the AI coach feedback and identify gaps.
5. Revise the solution using stronger technical and business logic.
6. Connect the scenario to deployment, monitoring, and continuous improvement.
7. Reorganize the conversation into a professional portfolio artifact.

9. Tools and Methods

- AI Data Challenge Scenarios Coach
- Random forest and gradient boosting concepts
- SMOTE and class weighting
- Precision, recall, F1-score, confusion matrix, AUC, and calibration
- Population Stability Index, Kolmogorov-Smirnov test, and Chi-square test
- Feature importance, threshold optimization, and validation testing
- Production monitoring and LLM operations concepts

10. Lessons Learned

This activity strengthened my understanding that machine learning success depends on more than choosing an algorithm. Data preparation, evaluation metrics, business costs, monitoring, and changing user populations all influence whether a model remains useful in practice.

The feedback process was especially valuable because it pushed me beyond broad recommendations. I learned to test whether missingness carries signal, quantify threshold decisions using business cost, measure drift statistically, and question whether retraining alone is enough when a new segment behaves differently.

The LLM discussion also showed that modern AI monitoring must evaluate behavior, factuality, safety, prompt changes, tool use, latency, and cost in addition to traditional predictive performance.

11. Value Proposition

This artifact demonstrates my ability to analyze realistic machine learning problems, explain technical tradeoffs clearly, incorporate feedback, and connect model decisions to business outcomes. It shows practical competency in preprocessing, cost-sensitive evaluation, drift diagnosis, model maintenance, and emerging AI operations. These capabilities are valuable for organizations that need reliable machine learning systems rather than isolated experimental models.

12. Conclusion

The inquiry-based exercise connected foundational machine learning concepts with real production challenges. Across the three scenarios, I developed solutions for missing data, severe class imbalance, and model drift, then extended the discussion to monitoring and maintaining both traditional machine learning and LLM-based systems.

The central lesson is that a model is not finished when it is deployed. Reliable AI requires continuous measurement, feedback, validation, and adaptation as data, customers, technology, and business requirements change.

Appendix: Original Inquiry Topics

- How should missing data be analyzed before imputation?
- When should a highly incomplete feature be retained or removed?
- How should class imbalance be handled when false negatives are much more costly?
- How can a prediction threshold be selected using expected business cost?
- How can data drift and concept drift be distinguished?
- Which statistical methods can quantify drift?
- When should a model be retrained, rebuilt, or segmented?
- What should be monitored after deployment?
- How does monitoring an LLM system differ from monitoring traditional ML?